Worksheet C

Number, Probability and Statistics

Edexcel International GCSE Mathematics Grades 6–8

Name: _____ Date: _____ 45 minutes 45 marks

Calculator allowed. Show all your working. Diagrams are not drawn accurately.

1 **[2]**

Show that the recurring decimal $0.\dot{2}\dot{7}$ can be written as $\frac{3}{11}$.

2 **[5]**

(a) After a 12% increase, the price of a jacket is £84. Work out the price before the increase. [2]

(b) £4000 is invested for 5 years at 3.5% per year compound interest. Work out the total value of the investment at the end of 5 years. Give your answer to the nearest penny. [3]

3 **[4]**

Give each answer in standard form.

(a) Work out $(3.6 \times 10^8) \times (5 \times 10^{-3})$ [2]

(b) Work out $\dfrac{4.8 \times 10^9}{1.6 \times 10^{-4}}$ [2]

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4 **[3]**

$a : b = 3 : 5$ and $b : c = 2 : 3$. Find $a : b : c$ in its simplest form.

5 **[5]**

(a) Solve $2^{3x-1} = 16$ [2]

(b) Simplify fully $\left(\frac{27y^3}{8x^6}\right)^{\frac{2}{3}}$ [3]

6 **[7]**

A bag contains 12 counters: 7 red and 5 blue. Two counters are taken at random without replacement.

(a) Work out the probability that at least one counter is blue. [3]

Another bag contains n counters, of which 4 are red. Two counters are taken at random without replacement. The probability that both are red is $\frac{6}{55}$.

(b) Show that $n^2 - n - 110 = 0$. [3]

(c) Hence find the value of n. [1]

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7 **[7]**

The table shows information about the heights, h cm, of 80 plants.

Height h (cm)	$0 < h \leq 10$	$10 < h \leq 30$	$30 < h \leq 40$	$40 < h \leq 60$
Frequency	12	30	24	14
Frequency density				

(a) Complete the frequency density row of the table. [2]

(b) Calculate an estimate for the mean height. [3]

(c) Estimate the number of plants taller than 35 cm. [2]

8 **[4]**

40 students were asked whether they study French (F) or Spanish (S). 22 study French, 15 study Spanish and 7 study both.

(a) How many students study neither language? [2]

(b) One of the students who studies Spanish is chosen at random. Find the probability that this student also studies French. [2]

9 **[3]**

A rectangle has length 12.5 cm and width 8.4 cm, both correct to 1 decimal place. Work out the upper bound for the area of the rectangle.

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10 **[5]**

A car is bought for £15 000. Its value falls by 14% each year.

(a) Show that its value after 3 years is £9540.84. [2]

(b) After how many whole years will its value first be less than £6000? [3]

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